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VISHWAKARMA UNIVERSITY, PUNE
University Grants Commission (UGC) Approved State Private University

Examination: Project-2			
Academic Year	2025-2026	Pattern	2024-2025
Faculty	Commerce and Management	Course Name	DevOps
Department	Computer Application	Course Code	MCA24401
Programme	MCA	Semester	IV
Division	A	Course Teacher	Dr. Bharati P. Kawade
Maximum Marks	25	Total No. of Pages	10
Project Start Date	05/01/2026	Project End Date	18/04/2026
Instructions to Candidates: - - Students need to strictly adhere to the project deadlines - Project content should be unique and plagiarism is strictly prohibited. - Submission of a project report in a prescribed format is mandatory.			

Project Title: - Quiz Game.

Project Objective:

The objective of this project is to design and develop an interactive **Quiz Game** application that allows users to answer questions, test their knowledge, and receive instant feedback. The project aims to make learning more engaging, enjoyable, and effective through a simple and user-friendly interface. It also helps in understanding the basic concepts of web development by building a complete working application.

Project Outcome:

1	Project Description	Marks	CO	BTL
2	Problem Definition / Identification	15	3	3,4
3	Implementations			
4	Output & Evidence			

*CO: Course Outcome, BTL: Bloom's Taxonomy Level

Course Outcomes (Write the Course Outcomes related to your project topic)

CO No.	Statement
1.	Many learners need a simple and engaging way to revise concepts, and a quiz game provides an interactive method to do so.
2.	Traditional study methods may not always keep users interested, so a quiz-based application can improve participation and focus.

3.	An online quiz game can make learning more enjoyable by combining education with a game-like experience.
4.	Students often need quick self-evaluation after studying, and a quiz game offers instant results and feedback.

Bloom's Taxonomy Level (BTL)

BTL No.	BTL	Statement
1	Remember	Recall facts and basic concepts
2	Understand	Explain ideas or concepts
3	Apply	Use information in new situations
4	Analyze	Draw connections among ideas
5	Evaluate	Justify a stand or decision
6	Create	Produce new or original work

Project Title: -
Quiz Game.

- **Project Definition**

- **Project Overview**

Objective:

The objective of this project is to design and develop an interactive **Quiz Game** application that allows users to answer questions, test their knowledge, and receive instant feedback. The project aims to make learning more engaging and enjoyable through a simple and user-friendly interface. It also helps in understanding basic software development concepts by creating a complete working application.

Scope:

The scope of this project includes creating a quiz game with multiple questions, answer options, scoring, and result display. The application can be expanded later with features such as timers, categories, difficulty levels, and leaderboards. It can run on web browsers and be made responsive for different devices. The project can also be improved by adding more topics and interactive elements.

- **Problem Statement**

Many students find it difficult to stay engaged with traditional study methods because they often do not provide instant feedback or interactive learning. Text-based learning alone may feel boring and less effective for revision. Therefore, a Quiz Game is needed to make learning fun, active, and self-assessment based. The application should allow users to attempt questions and check their performance immediately.

- **System Architecture**

Hardware Components

- Computer or Laptop
- Processor (Intel i3 or higher)
- Minimum 4 GB RAM
- Internet Connection

Software Components

- HTML
- CSS
- JavaScript
- Git & GitHub
- Web Browser (Chrome, Edge, etc.)

1.5 Hardware & Software Requirements

Hardware:

- System with at least 4 GB RAM
- Processor: Intel i3 or higher
- Hard Disk: Minimum 20 GB free space

Software:

- Operating System: Windows / Linux / Mac
- Code Editor: VS Code
- Web Browser: Google Chrome
- Version Control: Git & GitHub

• Implementation

• Implementation

The quiz game is developed using HTML, CSS, and JavaScript. HTML is used to create the structure of the game, CSS is used for styling, and JavaScript is used for question handling, scoring, and user interaction. The project code is written in a code editor such as VS Code. Git is used for version control, and GitHub is used to store and manage the project files.

• Code

```
import random
```

```
import time
```

```
quiz_question= [
```

```
{
```

```
    "question": "Which language is used to create web pages?",
```

```
"options":["Python", "HTML", "C++", "JAVA"],
"answer": "HTML"
},
{
"question": "Which data structure uses FIFO (First in first out)?",
"options":["Stack", "Queue", "Tree", "Graph"],
"answer": "Queue"
},
{
"question": "Who developed the Python programming language?",
"options": ["Guido van Rossum", "Elon Musk", "Bill Gates", "Sundar Pichai"],
"answer": "Guido van Rossum"
},
{
"question": "Which keyword is used to define a function in Python?",
"options": ["func", "define", "def", "function"],
"answer": "def"
},
{
"question": "What is the output of 5 + 3 * 2?",
"options": ["16", "13", "11", "10"],
"answer": "11"
},
{
"question": "Which of the following is not a programming language?",
"options": ["HTML", "Python", "Ruby", "C++"],
"answer": "HTML"
},
{
"question": "Which company owns the Android operating system?",
"options": ["Apple", "Google", "Microsoft", "Samsung"],
"answer": "Google"
},
{
"question": "What does CPU stand for?",
```

```

    "options": ["Central Process Unit", "Central Processing Unit", "Computer Processing Unit", "Central
Processor Utility"],
    "answer": "Central Processing Unit"
},
{
    "question": "Which planet is known as the Red Planet?",
    "options": ["Earth", "Venus", "Mars", "Jupiter"],
    "answer": "Mars"
},
{
    "question": "Which one is the largest social media platform?",
    "options": ["Twitter", "Instagram", "Facebook", "LinkedIn"],
    "answer": "Facebook"
}
]

```

```

def run_quiz():
    print(" 🎉 Welcome to the Fun Quiz Game!")
    time.sleep(1)
    name=input(" 🖱️ what's your name? ")
    print(f"\nHey {name}! 😊 Lets test your knowledge. \n")
    time.sleep(1)
    score=0
    total=len(quiz_question)

    random.shuffle(quiz_question)

    for i, q in enumerate(quiz_question, start=1):
        print(f"-----")
        print(f" 🎮 Question {i} of {total}")
        print(f"{q['question']}")
        for idx, option in enumerate(q["options"], start=1):
            print(f"{idx}. {option}")

    # "user input"

```

```
try:
    choice = int(input(" 🖱 Enter your answer (1-4): "))
    selected = q["options"][choice - 1]
except (ValueError, IndexError):
    print(" ⚠ Invalid input! Let's move to the next one.\n")
    time.sleep(1)
    continue

#check answers
if selected.lower() == q["answer"].lower():
    print(" ✅ Correct! Nice one! 🎯\n")
    score+=1
else:
    print(f" ❌ Oops! The correct answer is: {q['answer']}\n")
    time.sleep(1)

# Result summary
print("=====")
print(f" 🏁 Quiz finished, {name}!")
print(f" 📊 Your final score: {score}/{total}\n")

#feedback message
if score == total:
    print(" 🏆 Outstanding! You nailed every question!")
elif score >= total * 0.8:
    print(" 🤖 Great job! You clearly know your stuff.")
elif score >= total * 0.5:
    print(" 👍 Not bad! Keep practicing and you'll get even better.")
else:
    print(" 😓 Looks like you need a bit more practice. Don't give up!")
print("=====")
if __name__ == "__main__":
    run_quiz()
```

- **Output and Evidence**

- **Testing & Results Screenshots**

Results:

```
🎮 Welcome to the Fun Quiz Game!
👉 what's your name? vedanti

Hey vedanti! 🤖 Lets test your knowledge.

-----

🌸 Question 1 of 10
Which of the following is not a programming language?
1. HTML
2. Python
3. Ruby
4. C++
👉 Enter your answer (1-4): 1
✅ Correct! Nice one! 🎉

-----

🌸 Question 2 of 10
Which company owns the Android operating system?
1. Apple
2. Google
3. Microsoft
4. Samsung
👉 Enter your answer (1-4): 2
✅ Correct! Nice one! 🎉

-----

🌸 Question 3 of 10
Who developed the Python programming language?
1. Guido van Rossum
2. Elon Musk
3. Bill Gates
4. Sundar Pichai
👉 Enter your answer (1-4): 1
✅ Correct! Nice one! 🎉

-----

🌸 Question 4 of 10
What does CPU stand for?
1. Central Process Unit
2. Central Processing Unit
3. Computer Processing Unit
4. Central Processor Utility
👉 Enter your answer (1-4): 2
✅ Correct! Nice one! 🎉
```


✿ Question 5 of 10

What is the output of $5 + 3 * 2$?

1. 16
2. 13
3. 11
4. 10

👉 Enter your answer (1-4): 3

✅ Correct! Nice one! 🎉

✿ Question 6 of 10

Which language is used to create web pages?

1. Python
2. HTML
3. C++
4. JAVA

👉 Enter your answer (1-4): 2

✅ Correct! Nice one! 🎉

✿ Question 7 of 10

Which one is the largest social media platform?

1. Twitter
2. Instagram
3. Facebook
4. LinkedIn

👉 Enter your answer (1-4): 3

✅ Correct! Nice one! 🎉

✿ Question 8 of 10

Which keyword is used to define a function in Python?

1. func
2. define
3. def
4. function

👉 Enter your answer (1-4): 3

✅ Correct! Nice one! 🎉

✿ Question 9 of 10

Which data structure uses FIFO (First in first out)?

1. Stack
2. Queue
3. Tree
4. Graph

👉 Enter your answer (1-4): 2

✅ Correct! Nice one! 🎉

```
-----
🌟 Question 10 of 10
Which planet is known as the Red Planet?
1. Earth
2. Venus
3. Mars
4. Jupiter
👉 Enter your answer (1-4): 2
❌ Oops! The correct answer is: Mars

=====
🎉 Quiz finished, vedanti!
👤 Your final score: 9/10

👏 Great job! You clearly know your stuff.
=====
```

- **Applications**

- Interactive learning for students through quick question-and-answer practice.
- Self-assessment tool for users to test their knowledge on different topics.
- Classroom support for teachers to make revision more engaging.
- Exam preparation aid for competitive tests and subject-wise revision.
- A simple game-based platform that improves focus and participation.

- **Future Enhancements**

- Add a time for each question to make the game more challenging.
- Include multiple categories like science, math, GK, and technology.
- Add difficulty levels such as easy, medium, and hard.
- Store user scores and create a leaderboard.
- Improve the interface with animations, sounds, and better visuals.
- Add login functionality and save user progress.

- **Conclusion**

The Quiz Game project is a simple and effective application that combines learning with entertainment. It helps users test their knowledge, receive instant feedback, and stay engaged in a fun way. The project also provides practical experience in web development and can be improved further with advanced features.